

Towards Multimodal Large Language Models with Both Training and Inference Efficiency

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Abstract

Multimodal Large Language Models (MLLMs) mainly fall into two architectures, each involving a trade-off between training and inference efficiency: embedding space alignment (*e.g.* LLaVA series) is inefficient during inference, while cross-attention space alignment (*e.g.* Flamingo) is inefficient in training. A primary difference between them lies in whether each visual token attends to other tokens within the LLM backbones. To investigate whether this form of attention is essential for MLLMs, we propose NAEViT (No AttEntion from Visual Tokens), an attention mechanism that eliminates such interactions. Our pilot experiment shows that attention from visual tokens is highly redundant. Then, we introduce SAISA (Self-Attention Input Space Alignment), a novel architecture that enhances both training and inference efficiency. SAISA directly aligns visual features with the input spaces of NAEViT attention blocks, reducing computational overhead in both attention and FFNs. We conduct experiments on various baseline models, model sizes and training datasets. SAISA achieves superior performance compared to the baselines, while significantly reducing computational costs. Further ablation studies validate the effectiveness of SAISA across various LLMs and visual encoders.¹

1. Introduction

Multimodal Large Language Models (MLLMs) (OpenAI, 2023; Bai et al., 2023; 2025) have shown impressive capa-

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¹Code: <https://github.com/icip-cas/SAISA>

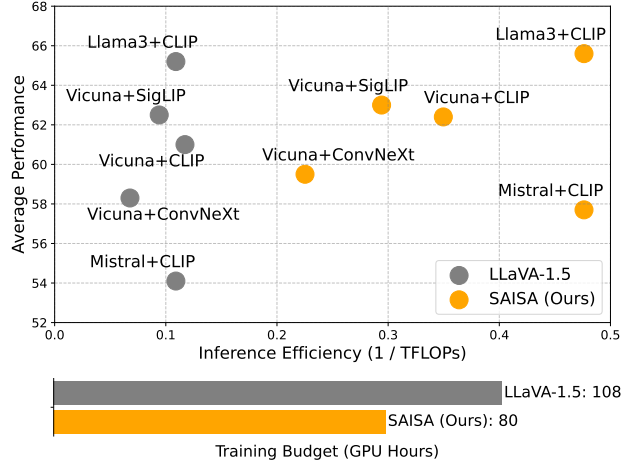


Figure 1. **Top:** Performance vs. inference efficiency based on various LLMs and visual encoders where Average Performance means an average of benchmark scores (MMMU, MMBench, MMBench-CN, POPE, GQA, SchienceQA and OK-VQA) and inference efficiency is the inverse of inference TFLOPs. **Bottom:** Training budget comparison based on the LLaVA-1.5-7B setting.

bilities in understanding and processing visual information. They typically build on pre-trained Large Language Models (LLMs) (Achiam et al., 2023; Team et al., 2024; Yang et al., 2025a) and align visual features with the LLMs. There are two primary architectures for aligning visual and text modalities: embedding space alignment and cross-attention space alignment. Embedding space alignment, *e.g.* LLaVA series (Liu et al., 2023; 2024a), introduces a projector to align visual features with the LLM embedding space and feeds the visual and text tokens into the LLM. Cross-attention space alignment, *e.g.* Flamingo (Alayrac et al., 2022), inserts cross-attention blocks and aligns visual features with the attention spaces of these blocks.

However, despite the promising performance of these MLLMs, they involve a trade-off between training and inference efficiency. On the one hand, MLLMs with embedding space alignment exhibit training efficiency, since they introduce only a small number of new parameters to the pre-trained LLMs. For example, LLaVA-1.5-7B is trained in 108 GPU hours from the text-only LLM backbone. How-

ever, this architecture significantly increases the number of input tokens, and the computational costs of self-attention grow quadratically with the number of tokens, leading to inefficiency during inference. On the other hand, MLLMs with cross-attention space alignment achieve inference efficiency, since they do not require unrolling visual tokens, but they are inefficient during training for introducing a large number of new parameters to the pre-trained LLM. In this paper, we take a step towards building MLLMs with efficiency during both training and inference.

In this paper, we first perform a thorough analysis of these two architectures, identifying key factors for building MLLMs with both training and inference efficiency. To optimize training efficiency, the key factor is minimizing the number of new parameters and employing modules in the pre-trained LLMs for interaction between visual and text modalities. For improving inference efficiency, the main focus is reducing the computational costs associated with visual tokens, particularly in attention blocks and feed-forward networks (FFNs). Since attention from visual tokens contributes significantly to the computational cost in self-attention blocks, we investigate whether this type of attention is truly essential for MLLMs. We introduce NAE-ViT, as shown in Figure 4, an attention mechanism where visual tokens do not function as queries to attend to other tokens. Our pilot experiment demonstrates that NAEViT outperforms the vanilla self-attention, indicating that attention from visual tokens is highly redundant.

Based on the findings above, we introduce SAISA (Self-Attention Input Space Alignment), an architecture for MLLMs with efficiency during both training and inference. As illustrated in Figure 2(c), SAISA employs NAE-ViT blocks for multimodal interaction and directly aligns the visual features with the input spaces of these blocks. SAISA not only reduces the computational overhead of self-attention blocks, but also significantly lowers the computational costs of FFNs by eliminating the need to apply FFNs to visual tokens. We validate its effectiveness on various experimental settings. As shown in Figure 1, SAISA outperforms the LLaVA architecture in terms of performance, training efficiency, and inference efficiency. SAISA reduces training budget by 26% and inference FLOPs by over 66%, while delivering superior performance.

Our contributions are three-fold.

- Based on our analysis of current MLLM architectures, we propose NAEViT to enhance efficiency of MLLMs, revealing the redundancy of attention in MLLMs.
- We introduce SAISA, an architecture for building MLLMs with both training and inference efficiency by eliminating the computational costs of attention from visual tokens, as well as FFNs on visual tokens.

- We validate the effectiveness of SAISA across various experimental settings, including different baseline models, model sizes and training datasets.

2. Analyzing MLLM Architectures

In this section, we analyze the two most common architectures to align visual features with the language model, and summarize key factors for building efficient MLLMs.

Embedding Space Alignment. As illustrated in Figure 2(a), models with this architecture introduce a projector to align visual tokens with the LLM embedding space. They concatenate visual tokens and text tokens, and then feed them into the LLM. Notable models with this architecture include LLaVA series (Liu et al., 2024a), Qwen series (Bai et al., 2023; 2025) and InternVL series (Zhu et al., 2025). These models introduce a small number of new parameters to the pre-trained LLMs, allowing training MLLMs from the LLMs with low budgets.

However, the concatenated token sequence leads to inference inefficiency. When the number of visual and text tokens is v and t respectively, the computational complexity of self-attention is $\mathcal{O}((v+t)^2)$. Typically, MLLMs use hundreds or thousands of visual tokens. For example, LLaVA-1.5 uses 576 visual tokens for a single image, and LLaVA-NeXT (Liu et al., 2024b) uses up to 2880. In contrast, the number of text tokens is much smaller in most tasks. The average numbers of text tokens in MMMU (Yue et al., 2024) and POPE (Li et al., 2023b) are 142 and 68, respectively. As a result, the attention from visual tokens, including vision-vision and vision-text attention, dominates the computational overhead. Moreover, since the FFNs have a large hidden layer dimension, applying FFNs to visual tokens also brings substantial computational costs.

Cross-Attention Space Alignment. As illustrated in Figure 2(b), models with this architecture insert cross-attention blocks and FFNs into the language model, and align visual features with the attention spaces of these cross-attention blocks. Notable models with this architecture include Flamingo (Alayrac et al., 2022), OpenFlamingo (Awadalla et al., 2023) and Otter (Li et al., 2023a). In these models, the attention operation consists of only two components, the interaction between text and visual modalities with complexity $\mathcal{O}(vt)$ in the cross-attention blocks, and the attention among text tokens with complexity $\mathcal{O}(t^2)$ in the self-attention blocks. Compared with embedding space alignment, there is no attention from visual features in the language model. By not executing this type of attention and not applying FFNs to visual tokens, these models are more efficient during inference.

However, the inserted cross-attention blocks and FFNs introduce a large number of new parameters to the pre-trained

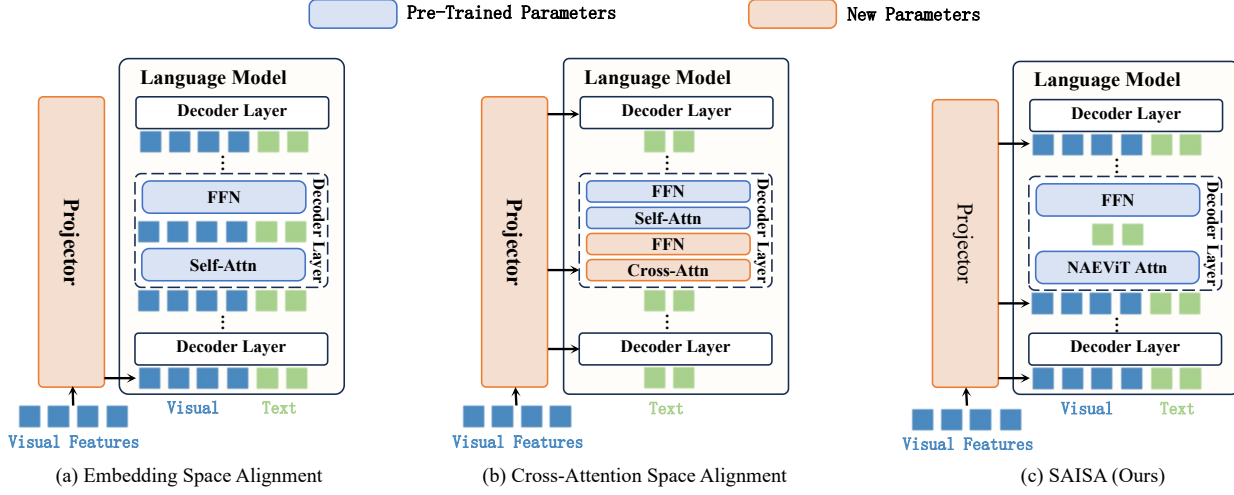


Figure 2. Overview of SAISA and mainstream architectures to align visual features with language models. (a) Aligning visual features with the embedding space of language models is inefficient during inference, *e.g.* LLaVA. (b) Aligning visual features with the attention spaces of new cross-attention blocks is inefficient during training, *e.g.* Flamingo. (c) SAISA aligns visual features with the attention input spaces of language models, achieving efficiency during both training and inference.

language model. As a result, training an MLLM with this architecture requires a large amount of data. For example, OpenFlamingo-9B adds 2 billion parameters to Llama-7B (Touvron et al., 2023a), and requires training data with 180M samples. In terms of model capabilities, previous work (Dai et al., 2024) finds that models utilizing this architecture perform worse than those using embedding space alignment when trained on the same data.

Based on the analyses above, we summarize the key factors for building efficient MLLMs as follows: (1) Reducing the number of new parameters and employing pre-trained modules for multimodal interaction lead to efficiency during training. (2) Reducing computations related to visual tokens, including those in attention blocks and FFNs, leads to efficiency during inference.

3. No Attention From Visual Tokens

In this section, we propose NAEViT attention and perform a pilot experiment to investigate whether attention from visual tokens is essential for MLLMs.

3.1. Preliminary

Vanilla Self-Attention. In a self-attention block, the input visual-text token sequence is formed as $X = [V, T] \in \mathbb{R}^{(v+t) \times h}$, where $[\cdot, \cdot]$ denotes concatenation along the sequence dimension. To derive the query, key, and value representations, three linear layers are applied to obtain X_q , X_k , and X_v , respectively:

$$X_q = [V, T]W_Q, X_k = [V, T]W_K, X_v = [V, T]W_V \quad (1)$$

Then, the attention operation is executed as:

$$\text{Attention}(X) = \text{softmax} \left(\frac{X_q X_k^T}{\sqrt{d}} \right) X_v \in \mathbb{R}^{(v+t) \times h} \quad (2)$$

Typically, a causal attention mask is applied and queries can only attend to keys preceding them in the sequence. The outputs are multiplied by another linear layer W_O to update the hidden states through a residual connection:

$$\text{SA}(X) = \text{Attention}(X)W_O + X. \quad (3)$$

3.2. Attention Redundancy Analyses

As analyzed in Section 2, embedding space alignment achieves superior performance and training efficiency. However, it is inefficient during inference, attributable to the self-attention from visual tokens. In contrast, in cross-attention space alignment, visual tokens do not attention to other tokens in the language model backbone, leading to inference efficiency. A question arises here: **Is attention from visual tokens essential for MLLMs?**

To quantify the contribution of attention from visual tokens, we utilize KL divergence as a metric, inspired by Lin et al. (2024); Yuan et al. (2025b); Men et al. (2025). For an attention block B , we eliminate B 's transformations on different types of tokens, respectively, *i.e.* keeping the hidden states of the investigated tokens unchanged in B , and compute the KL divergence between the resulting model's output logits and those of the original model. When the investigated tokens are visual tokens, eliminating such transformations is equivalent to removing attention from visual tokens. A lower KL divergence indicates that removing such transfor-

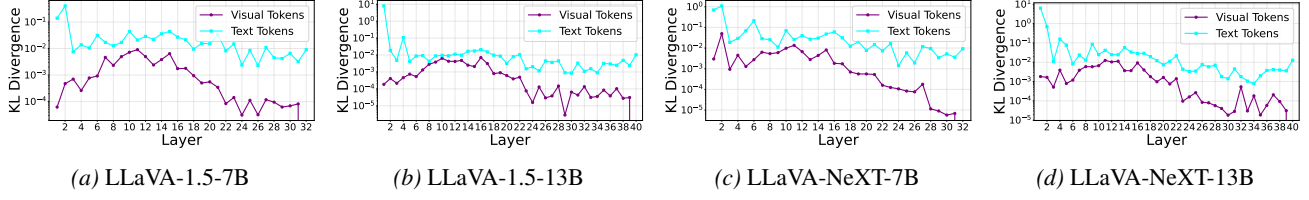


Figure 3. Self-attention blocks' contributions to MLLMs' output, measured by KL ivergence.

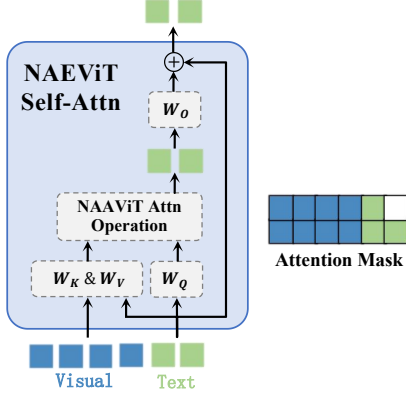


Figure 4. NAEViT self-attention block. It uses only text tokens as queries. They attend to visual tokens and text tokens preceding them. Visual tokens are not updated in this block.

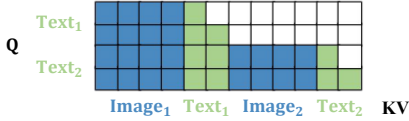


Figure 5. Attention mask for interleaved image-text.

mations in the investigated block has minimal impact on the model's output.

As shown in Figure 3, the attention blocks' transformations on visual tokens consistently contribute less than those on text tokens. This observation suggests that attention from visual tokens are highly redundant, inspiring us to introduce NAEViT in the following subsection.

3.3. NAEViT

We propose NAEViT (No AttEntion from Visual Tokens), which eliminates attention from visual tokens. We illustrate the architecture of NAEViT in Figure 4 (Top). Specifically, for the visual-text token sequence $X = [V, T] \in \mathbb{R}^{(v+t) \times h}$, queries X_q , keys X_k and values X_v are obtained as:

$$X_q = TW_Q, X_k = [V, T]W_K, X_v = [V, T]W_V \quad (4)$$

The attention operation is executed as Equation 2, but with NAEViT attention mask, where the queries can attend to visual tokens and text tokens preceding them.

When the number of visual and text tokens is v and t respec-

tively, the vanilla self-attention exhibits a computational complexity of $\mathcal{O}((v+t)^2)$ for the attention operation in Equation 2. By eliminating the attention from visual tokens, NAEViT reduces the complexity to $\mathcal{O}(t(v+t))$. Furthermore, since NAEViT reduces the query length from $v+t$ to t , it also significantly reduces the computational costs associated with matrices W_Q and W_O in attention blocks.

3.4. Pilot Experiment

We train a model under the same configurations as LLaVA-1.5-7B, but replace the vanilla self-attention with NAEViT.

In Table 1, we compare NAEViT and vanilla self-attention on multiple MLLM benchmarks. Among them, MMMU contains interleaved text-image inputs. For these inputs, the NAEViT attention mask is illustrated in Figure 4 (Bottom). Despite eliminating the attention from visual tokens, the model employing NAEViT outperforms the one using the vanilla self-attention. Given that NAEViT substantially reduces computational overhead, it offers a favorable balance between performance and efficiency.

In conclusion, attention from visual tokens is highly redundant for building MLLMs. In the following section, we introduce SAISA for efficient MLLMs based on NAEViT. We provide the direct efficiency and performance comparison between models with NAEViT and SAISA in the appendix.

4. SAISA

4.1. Architecture

As mentioned earlier, besides the attention operation, another factor that contributes to inference inefficiency is applying FFNs to visual tokens. Based on NAEViT, which eliminates the attention from visual tokens, we propose SAISA (Self-Attention Input Space Alignment), an architecture for further enhancing MLLM efficiency. In SAISA, we also eliminate FFNs' computations on visual tokens.

We illustrate the SAISA architecture in Figure 2(c). SAISA contains a visual encoder to extract visual features, a projector, and an LLM. Each layer of the LLM consists of a self-attention block and an FFN. We utilize NAEViT in the self-attention blocks. The purpose of the projector is to directly align the visual features with the input spaces of

Table 1. Effects of attention mechanisms.

Method	MMMU VAL	MMBench EN CN	POPE	GQA	SQA IMG	OK- VQA	Text VQA	OCR Bench	Avg.
Vanilla Self-Attention	35.7	64.3 58.3	86.8	62.0	66.8	53.4	58.2	31.3	57.4
NAEViT (Ours)	36.0	64.9 58.4	86.9	62.4	68.7	56.0	58.1	31.3	58.1

different self-attention blocks in the LLM.

Specifically, we assume n is the number of layers in the LLM, h is the hidden size of the LLM, d is the dimension of visual features, v is the number of visual tokens, and t is the number of text tokens. For an input image I , we first employ the visual encoder VE to extract visual features:

$$Z = \text{VE}(I) \in \mathbb{R}^{v \times d} \quad (5)$$

Then, we use the projector P to directly align the visual features with the input spaces of different NAEViT self-attention blocks:

$$V = P(Z) \in \mathbb{R}^{n \times v \times h} \quad (6)$$

For the i -th NAEViT self-attention block NAEViT $_i$, we input the aligned visual features $V_i \in \mathbb{R}^{v \times h}$ and hidden states of the text tokens $T_i \in \mathbb{R}^{t \times h}$:

$$H_i = \text{NAEViT}_i(V_i, T_i) \in \mathbb{R}^{t \times h}, i = 1, 2, 3, \dots, n \quad (7)$$

Notably, the NAEViT attention block only outputs the text tokens' hidden states H_i for the subsequent FFN, and we apply the FFN to update the text tokens:

$$T_{i+1} = \text{FFN}_i(H_i) \in \mathbb{R}^{t \times h}, i = 1, 2, 3, \dots, n. \quad (8)$$

SAISA utilizes T_{n+1} for the next-token prediction.

4.2. Projector

Since each LLM layer operates in distinct attention spaces, the projector must flexibly align the visual features with each of the spaces. For simplicity, we employ distinct two-layer MLPs for each layer of the LLM. When the LLM has n layers, the projector contains n MLPs. We set the intermediate size of each MLP to be the same as the LLM hidden size.

Specifically, for the i -th layer of the LLM,

$$V_i = \text{MLP}_i(Z) = \varphi(ZW_{i,1})W_{i,2}, i = 1, 2, 3, \dots, n \quad (9)$$

where $Z \in \mathbb{R}^{v \times d}$ is the visual features from the visual encoder, φ is the activation function like GELU (Hendrycks & Gimpel, 2016), $W_{i,1} \in \mathbb{R}^{d \times h}$ and $W_{i,2} \in \mathbb{R}^{h \times h}$ are the weight matrices of the two fully connected layers.

4.3. Training Procedure

Pre-training. The objective of pre-training is to transform an LLM into an MLLM with a foundational comprehension of visual information, providing an initialization for the multimodal fine-tuning stage. Following previous work (Liu et al., 2024a; Shi et al., 2025), only the multimodal projector is trainable during this stage. To improve training efficiency, we further reduce the number of trainable parameters in this stage. Specifically, we train a shared MLP for all layers of the LLM.

Fine-tuning. At the beginning of this stage, we replicate the pre-trained shared MLP N times to initialize the MLPs in the projector, where N denotes the number of layers in the LLM backbone. Both the LLM and the projector are trainable during this stage.

4.4. Comparison of Computational Cost

We focus on the computations of the LLM and the projector, because those of the visual encoder are identical in this comparison. For LLM with n layers, we assume h is the hidden state size, m is the intermediate size of the FFNs, t is the number of text tokens, and v is the number of visual tokens. To comprehensively consider LLMs with and without grouped query attention (GQA) (Ainslie et al., 2023), we assume that k is the output dimension of key/value matrices. For the projector, we assume that d is the dimension of the input visual features. The FLOPs of LLaVA-1.5 are $2n(t+v)h(2h+3m+2k)+4n(t+v)^2h+2vhd+2vh^2$. For SAISA, visual tokens are multiplied only by the key and value matrices, and the query sequence length is t in the attention operation. Therefore, the FLOPs are $2nth(2h+3m+2k)+4nvhk+4nt(t+v)h+2nvhd+2nvh^2$. Details of the FLOPs calculation are in Appendix A.

5. Experiments

5.1. Experiment Settings

To validate the effectiveness of SAISA, we conduct a range of experiments under different settings. To make fair comparisons, we select multiple MLLMs with open source training data as baselines, which enable us to train the corresponding SAISA models with the same data and settings. For single-image tasks, we compare SAISA with LLaVA-1.5 (Liu et al., 2024a) and EAGLE (Shi et al., 2025). For

Table 2. Efficiency and performance comparison across settings, including different (1) MLLM architectures and training datasets, (2) model scales.

Method	#Data	FLOPs (T)↓	MMMU VAL	MMBench EN	MMBench CN	POPE	GQA	SQA IMG	OK- VQA	Text VQA	OCR Bench	Avg.
LLaVA-1.5-7B	1.2M	8.53	35.7	64.3	58.3	86.8	62.0	66.8	53.4	58.2	31.3	57.4
w/ SAISA (Ours)	1.2M	2.86	36.9	65.7	59.0	87.2	60.9	70.1	56.8	56.8	31.1	58.3
LLaVA-1.5-13B	1.2M	16.61	35.3	67.7	63.6	87.1	63.3	71.6	58.2	61.3	33.6	60.2
w/ SAISA (Ours)	1.2M	5.52	35.9	68.1	64.1	87.3	62.9	71.7	58.1	61.2	33.6	60.3
EAGLE-X5-7B	2.4M	14.79	36.3	68.4	61.5	88.5	64.9	70.0	58.8	71.7	53.8	63.8
w/ SAISA (Ours)	2.4M	5.54	37.1	68.8	62.1	88.5	64.8	71.0	58.6	71.9	54.1	64.1

Table 3. Comparison of various methods for MLLM efficiency. The best results among efficient methods are highlighted in bold.

Method	FLOPs (T)↓	MMMU VAL	MMBench EN	MMBench CN	POPE	GQA	SQA IMG	OK- VQA	Text VQA	OCR Bench
LLaVA-1.5-7B	8.53	35.7	64.3	58.3	86.8	62.0	66.8	53.4	58.2	31.3
w/ FastV (ECCV24)	4.90	35.8	64.3	56.8	82.5	60.2	67.8	52.9	56.4	30.3
w/ VTW (AAAI25)	4.70	36.3	64.0	58.5	86.0	55.1	69.1	51.0	51.9	5.0
w/ TopV (CVPR25)	4.35	35.1	59.7	-	84.2	-	69.6	52.6	-	31.0
w/ SparseVLM (ICML25)	2.84	33.8	60.0	51.1	85.3	59.5	67.1	-	56.1	-
w/ PAR (ICCV25)	4.00	-	60.2	50.7	73.6	-	69.6	-	-	-
w/ VisiPruner (EMNLP25)	3.93	-	62.0	-	84.4	60.3	66.7	-	55.2	-
w/ CDPPruner (NeurIPS25)	2.51	-	63.1	55.0	87.7	59.9	69.0	-	56.2	-
w/ SAISA (Ours)	2.86	36.9	65.7	59.0	87.2	60.9	70.1	56.8	56.8	31.1
LLaVA-NeXT-7B	42.67	36.4	67.1	60.6	86.1	64.1	70.1	44.2	61.3	52.4
w/ FastV (ECCV24)	22.04	35.1	66.3	58.5	85.5	63.0	69.0	43.0	58.6	50.8
w/ CDPPruner (NeurIPS25)	9.40	-	66.3	57.5	87.3	62.6	67.9	-	58.4	-
w/ SAISA (Ours)	10.98	36.1	66.9	57.8	87.6	63.1	69.3	43.8	59.4	51.3

multi-image and video tasks, we compare SAISA with Mantis (Jiang et al., 2024), an MLLM tailored for these scenarios with open source data.

5.2. Main Results

For single-image tasks, we conduct experiments on a range of benchmarks, including: (1) comprehensive MLLM benchmarks, such as MMMU (Yue et al., 2024), MM-Bench (Liu et al., 2024c) and MMBench-CN (Liu et al., 2024c); (2) hallucination benchmark such as POPE (Li et al., 2023b); (3) general visual question answering benchmarks such as GQA (Hudson & Manning, 2019) and ScienceQA IMG (Lu et al., 2022); (4) fine-grained visual question answering benchmarks such as OK-VQA (Marino et al., 2019), TextVQA (Singh et al., 2019) and OCRBench (Liu et al., 2024d), OK-VQA requires fine-grained image understanding and spatial understanding, while TextVQA and OCRBench are OCR-related benchmark; (5) vision-centric MLLM benchmarks such as MMVP (Tong et al., 2024) and CV-Bench (Tong et al., 2025).

Table 2 shows the comparison on the comprehensive benchmarks. We report the TFLOPs of processing a single image

Table 4. Results on multi-image (NLVR2 and Mantis-Eval) and video (LongVideoBench, LVBench and NExTQA) scenarios, based on Mantis (Jiang et al., 2024), an MLLM tailored for these tasks with fully open source training data.

Method	NLVR2	Mantis -Eval	LongVideo Bench	LV Bench	NExTQA
Mantis-CLIP	84.7	53.0	45.1	35.5	72.7
w/ SAISA	84.8	54.6	45.7	35.1	73.2

when the number of text tokens is 64. We also provide the comparison on vision-centric benchmarks (MMVP and CV-Bench) in the appendix. SAISA achieves the better overall performance compared to the baselines, and strikes a favorable balance between performance and efficiency.

For multi-image tasks, we conduct experiments on NLVR2 (Suh et al., 2019) and Mantis-Eval (Jiang et al., 2024). For video tasks, we conduct evaluation on LongVideoBench (Wu et al., 2024), LVBench (Wang et al., 2025) and NExTQA (Xiao et al., 2021). As shown in Table 4, SAISA consistently improves performance on most evaluated tasks, demonstrating its effectiveness in both multi-image reasoning and video understanding scenarios.

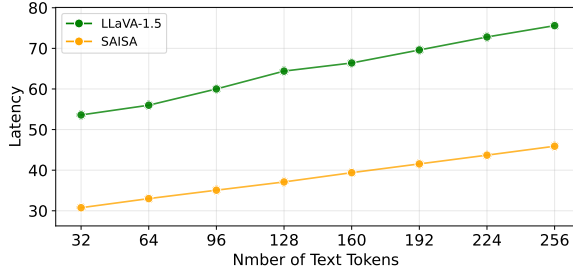


Figure 6. Inference latency, with a single image input.

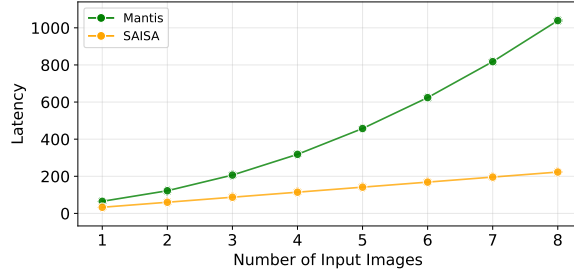


Figure 7. Inference latency, with different numbers of images.

Table 5. Ablation on LLMs and Visual Encoders, based on the LLaVA-1.5 setting. Here, “SigLIP” = SigLIP-ViT-SO400M/14-384, “ConvNeXt” = OpenCLIP-ConvNeXt-XXL-1024, and “CLIP” = CLIP-ViT-L/14-336. *Both LLaVA-1.5 and SAISA models on Mistral exhibit low performance on OK-VQA, because they respond to most questions in this benchmark with “Unanswerable”.

Method	LLM	Visual Encoder	FLOPs (T)↓	MMMU VAL	MMBench EN	MMBench CN	POPE	GQA	SQA IMG	OK-VQA	Avg.
LLaVA-1.5	Vicuna	SigLIP	10.63	36.6	66.2	58.9	86.5	62.5	70.5	56.4	62.5
w/ SAISA (Ours)	Vicuna	SigLIP	3.40	37.4	67.5	60.7	87.0	62.9	70.0	55.8	63.0
LLaVA-1.5	Vicuna	ConvNeXt	14.76	34.6	56.6	49.6	88.2	61.1	66.4	51.4	58.3
w/ SAISA (Ours)	Vicuna	ConvNeXt	4.44	35.1	61.1	54.9	87.0	57.7	66.5	54.4	59.5
LLaVA-1.5	Vicuna	CLIP	8.53	35.7	64.3	58.3	86.8	62.0	66.8	53.4	61.0
w/ SAISA (Ours)	Vicuna	CLIP	2.86	36.9	65.7	59.0	87.2	60.9	70.1	56.8	62.4
LLaVA-1.5	Mistral	CLIP	9.17	34.8	65.9	54.9	87.2	62.0	71.6	2.5*	54.1
w/ SAISA (Ours)	Mistral	CLIP	2.10	35.9	67.5	57.5	86.9	61.2	71.2	23.9*	57.7
LLaVA-1.5	Llama3	CLIP	9.17	36.8	70.4	64.2	87.2	63.5	73.3	61.2	65.2
w/ SAISA (Ours)	Llama3	CLIP	2.10	38.3	71.3	65.2	86.8	61.8	74.4	60.7	65.6

Moreover, we also compare SAISA with homogeneous methods for improving MLLM efficiency, such as FastV (Chen et al., 2024), VTW (Lin et al., 2024), TopV (Yang et al., 2025b), SparseVLM (Zhang et al., 2024), PAR (Suo et al., 2025), VisiPruner (Fan et al., 2025) and CDPPruner (Zhang et al., 2025). As shown in Table 3, SAISA achieves the best overall performance under a comparable or even lower computational budget, consistently outperforming all competing methods on most benchmarks. In contrast, existing approaches often exhibit a clear trade-off between efficiency and accuracy. Overall, the comparison demonstrates that SAISA provides a more robust and general solution for efficient MLLMs, achieving substantial FLOPs reduction without sacrificing, and in many cases even improving, model performance across diverse tasks.

In addition to performance, we also conduct inference latency tests on a single A100 GPU, measuring Time To First Token under different settings. In Figure 6, we illustrate the latency of LLaVA-1.5-7B and the corresponding SAISA model using a single image input while varying the number of text tokens. In Figure 7, we report the latency under different numbers of image inputs. With more images as input, SAISA achieves greater speedups.

Notably, SAISA inherently supports flexible multi-turn con-

versations. We include multi-turn results in Table 10.

5.3. Ablation Study

In this subsection, we conduct ablations on LLM backbones, visual encoders, pre-training strategies and projector designs. The ablations are based on the LLaVA-1.5-7B configuration.

Ablation on LLMs and Visual Encoders. As presented in Table 5, we perform multiple ablations on both LLM backbones and visual encoders to validate the effectiveness of SAISA. We tune a set of SAISA models using a variety of LLMs and visual encoders. The ablated LLMs include Vicuna-7B (Chiang et al., 2023) and two LLMs using grouped query attention (GQA) (Ainslie et al., 2023), such as Mistral-7B (Jiang et al., 2023) and Llama3-8B (Grattafiori et al., 2024). The ablated visual encoders include two ViT-based (Dosovitskiy et al., 2020) visual backbones such as CLIP-ViT-L/14-336 (Radford et al., 2021) and SigLIP-ViT-SO400M/14-384 (Zhai et al., 2023), and a ConvNeXt-based (Liu et al., 2022) visual encoder such as ConvNeXt-XXL-1024 from OpenCLIP (Ilharco et al., 2021; Schuhmann et al., 2022). The experimental results demonstrate that SAISA consistently achieves superior performance across different LLMs and visual encoders, while

Table 6. Ablation on Pre-training Strategies.

Pre-trained Parameters	MMMU VAL	MMBench EN CN	POPE	SQA IMG	OK- VQA	
Full Projector	34.8	59.2	51.1	85.6	67.8	53.1
Shared MLP	36.9	65.7	59.0	87.2	70.1	56.8

Table 7. Ablation on Projector Designs.

Projector	MMMU VAL	MMBench EN	CN	POPE	SQA IMG	OK- VQA
Linear Layers	35.7	65.3	56.6	85.8	69.2	53.6
Resamplers	35.6	59.5	50.5	83.5	69.1	53.4
MLPs	36.9	65.7	59.0	87.2	70.1	56.8

dramatically reducing computational costs.

Ablation on Pre-training Strategies. As shown in Table 6, we conduct an ablation to investigate the effects of SAISA’s pre-training strategies. We tune a SAISA model where the full projector is tunable during pre-training. With more randomly initialized parameters, we observe a performance drop when pre-training the full projector. We attribute this drop to the small amount of pre-training data with only 558k samples. This ablation study demonstrates the effectiveness of our pre-training strategy, which provides a robust initialization for the subsequent fine-tuning stage.

Ablation on Projector Designs. Previous studies find that replacing linear projection with MLP projection improves MLLM performance (Liu et al., 2024a). We conduct an experiment to investigate the impact of projectors in SAISA. We replace each MLP in the projector with a linear layer or a resampler (Jaegle et al., 2021). We set the output token number of the resampler projector to 256. Table 7 shows that the model with the MLP projector performs better than the model with other projectors, including the linear projector, which is consistent with the finding of the previous study (Liu et al., 2024a). Furthermore, the linear projector performs better than the resampler projector, because the latter down-samples visual tokens and overlooks fine-grained visual information such as spatial information.

6. Related Work

6.1. Multimodal Large Language Models

Large Language Models (LLMs) (Comanici et al., 2025; Touvron et al., 2023a; Bai et al., 2025) have demonstrated strong capabilities in solving complex tasks (Mo et al., 2025; Yuan et al., 2025a; Chen et al., 2025; Zheng et al., 2026; Touvron et al., 2023b). Multimodal Large Language Models (MLLMs) (OpenAI, 2023; Yuan et al., 2026; Bai et al., 2025) are typically built on LLMs by aligning visual features generated by visual encoders (Radford et al., 2021; Zhai et al.,

2023; Ilharco et al., 2021; Liu et al., 2022) with the LLMs. There are two most common architectures for this purpose, embedding space alignment and cross-attention space alignment. For embedding space alignment (Liu et al., 2023; 2024a; Bai et al., 2023; 2025; Zhu et al., 2025), MLLMs align visual features with the text token embedding space via a projector and concatenate the visual and text tokens as the LLM input. These models exhibit efficiency during training, but suffer from inference inefficiency because of the long token sequence. For cross-attention space alignment (Alayrac et al., 2022; LAION, 2023; Awadalla et al., 2023), MLLMs introduce new cross-attention blocks for the interaction between text and visual modalities, and align the visual features with the cross-attention spaces. These models achieve efficiency during inference, but require a substantial amount of data to train the new parameters. In this paper, we propose SAISA, an architecture for building MLLMs with efficiency during both training and inference.

6.2. Visual Redundancy and Efficiency Optimization

Some previous work has discussed redundancy in ViT (Bolya et al., 2022) and MLLMs (Li et al., 2025), and proposed methods to reduce computational costs of MLLMs. Notably, Shukor & Cord (2024) expose frozen LLMs to multiple modalities, finds that visual tokens change slightly throughout LLMs, and proposes to skip computations in some FFNs. Chen et al. (2024) and Lin et al. (2024) reveal the redundancy of visual tokens and prune visual tokens in deep layers, while Shang et al. (2025) and Yang et al. (2025d) prune and merge visual tokens produced by the vision encoder. Yang et al. (2025c) probe visual tokens and explain visual redundancy through contexts. Ye et al. (2024) incorporate cross-attention operation in parallel with self-attention, and utilizes adaptive gates for modality fusing. Ma et al. (2024) introduce aligners to update visual tokens, but the aligners still involve substantial computational overhead. Unlike previous work, we propose SAISA, a new architecture to build and train efficient MLLMs. Instead of reducing visual token number, SAISA achieves efficiency while maintaining fine-grained visual understanding by preserving the original number of visual tokens.

7. Conclusion

In this paper, we take a step towards developing MLLMs with efficiency during both training and inference. To achieve this, we conduct a study of current MLLM architectures and find the key factors for building efficient MLLMs. By integrating these factors and gradually reducing redundant computations, we propose SAISA, an effective and efficient architecture for MLLMs. SAISA demonstrates the ability to dramatically reduce the computational costs of MLLMs without compromising their capabilities.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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Table 8. Comparison between NAEViT and SAISA

Method	FLOPs (T)↓	MMMU VAL	MMBench EN CN	POPE	GQA	SQA IMG	OK- VQA	Text VQA	OCR Bench	Avg.
LLaVA-1.5-7B	8.53	35.7	64.3 58.3	86.8	62.0	66.8	53.4	58.2	31.3	57.4
w/ NAEViT	7.10	36.0	64.9 58.4	86.9	62.4	68.7	56.0	58.1	31.3	58.1
w/ SAISA	2.86	36.9	65.7 59.0	87.2	60.9	70.1	56.8	56.8	31.1	58.3

A. Analysis of Computational Overhead

In this section, we provide details of the FLOPs calculation. We consider the computations of the LLM and the projector, as the computations of the visual encoder are identical in comparison. We consider the computations of the self-attention blocks and the FFNs in each layer of the LLM.

We denote the hidden state size of the LLM backbone as h , and the output dimension of the W_K , W_V matrices is k . When the token sequence has l tokens, the FLOPs of a vanilla self-attention block contain those of the W_Q , W_K , W_V and W_O matrices, as well as the self-attention operation. The FLOPs of each of W_K , W_V are $2lhk$, and those of each of W_Q and W_O are $2lh^2$, and the FLOPs of the self-attention operation are $4l^2h$. The overall FLOPs of the vanilla self-attention block are $4lh^2 + 4lhk + 4l^2h$. For an feed-forward network (FFN), we assume that their intermediate size is m , and the FLOPs are $6lhm$. When the LLM backbone has n layers, the FLOPs are $2nlh(2h + 3m + 2k) + 4nl^2h$.

We use v to denote the number of visual tokens, and t to denote the number of text tokens. For LLaVA-1.5, the visual and text tokens are concatenated and then fed into the LLM backbone, resulting in $v + t$ tokens. As a result, the FLOPs of the LLM backbone are $2n(v + t)h(2h + 3m + 2k) + 4(v + t)l^2h$. For the 2-layer MLP projector, where the hidden layer size is set to h , the FLOPs of the first layer are $2vhd$, and those of the second layer are $2vh^2$. The overall FLOPs of LLaVA-1.5 are $2n(v + t)h(2h + 3m + 2k) + 4(v + t)l^2h$.

For SAISA, a NAEViT self-attention block also contains W_Q , W_K , W_V , W_O and the self-attention operation. In contrast to the vanilla self-attention block, the W_Q and W_O matrices are only applied to text tokens in SAISA, and the attention operation from visual tokens are omitted. As a result, the overall FLOPs of a NAEViT self-attention block are $2th(2h + 2k) + 4vkh + 4t(t + v)h$. The FFNs in SAISA are only applied to text tokens, resulting in FLOPs $6thm$. With respect to the projector, which contains n 2-layer MLPs, the FLOPs are $2nvhd + 2nvh^2$. The overall FLOPs of SAISA are $2nth(2h + 3m + 2k) + 4nvkh + 4nt(t + v)h + 2nvhd + 2nvh^2$.

B. Comparison Between NAEViT and SAISA

We present a direct efficiency and performance comparison between the models with NAEViT and SAISA as in Table 8. NAEViT reduces FLOPs by 17%, while SAISA achieves a 66% reduction. This implies that the removal of FFNs accounts for 49% reduction in FLOPs, representing a substantial portion of the overall efficiency gains.

However, we want to note that although the removal of FFNs contributes the most to the efficiency gains, the effectiveness of NAEViT is the foundation that enables us to remove FFNs on visual tokens. After we apply NAEViT, a visual tokens only passes through multiple FFNs (MLPs) in the LLM backbone, and it doesn't interact with other tokens. Then we can simplify the transformation of multiple FFNs to a single 2-layer MLP, which results in SAISA. In contrast, when the transformation involves interactions with other tokens (as modeled by attention mechanism), it cannot be simplified to a simple MLP.

Regarding the performance comparison, we find that SAISA achieves better overall performance than using NAEViT alone. We attribute this improvement to the following reason: when using NAEViT alone, we apply FFNs to both visual and text tokens. Since these FFN parameters in the LLM backbone are pre-trained on a large scale of text data, they are optimized for processing text tokens. Although we then train them on both visual and text tokens, as the two modalities demonstrate a clear modality gap in the embedding space (Huang et al., 2025), and the scale of optimized visual tokens is typically smaller than text tokens, these parameters cannot effectively handle both modalities. We believe this imbalance is also a reason of the "language-prior" phenomenon. In contrast, SAISA uses other lightweight MLPs to process visual tokens, and FFNs are optimized to process only text tokens.

Table 9. Performance on vision-centric benchmarks.

Method	Inference TFLOPs↓	MMVP	CV-Bench	
			2D	3D
LLaVA-1.5	8.53	24.7	56.6	59.5
SAISA (Ours)	2.86	26.0	56.2	59.8

Table 10. SAISA can dynamically adjust response granularity across multiple dialogue rounds.

Multi-turn Conversation:

User	What is unusual about this image? Answer briefly with a single sentence.
LLaVA-1.5	A man is hanging clothes on a clothesline attached to a car.
SAISA	A man is ironing clothes while standing on the back of a taxi.
User	Give a more detailed explanation.
LLaVA-1.5	A man is hanging clothes on a clothesline attached to a car, which is an unusual sight as people typically hang clothes on a clothesline at home or in a designated area.
SAISA	The unusual aspect of this image is that a man is ironing clothes while standing on the back of a taxi. This is not a typical scene, as ironing clothes is usually done indoors, and standing on the back of a moving vehicle is not a safe or practical way to do it. The man’s actions may be a stunt or a publicity stunt, but it is certainly an unconventional and potentially dangerous situation.

C. Vision-Centric Benchmarks

In this section, we provide the comparison on vision-centric benchmarks. The results are shown in Table 9.

D. Qualitative Results

In this section, we utilize the qualitative examples in the original LLaVA-1.5 paper to comprehensively demonstrate the abilities of SAISA. The examples, which cover a diverse range of scenarios and challenges, are shown in Table 12, 11, 13 and 14. For comparison, we prompt queries in these examples to LLaVA-1.5 and include its response in most examples. Our SAISA, trained on the same data as LLaVA-1.5, demonstrates similar performance across various unseen response format prompts, while exhibiting superior training and inference efficiency. It can support flexible multi-turn conversations (Table 10) and accurately identify factual inaccuracies in questions and provide more effective and reasonable responses to complex queries (Table 12). SAISA also generates fine-grained prompts for Stable Diffusion models in a specified format (Table 11). Furthermore, it can offer responses with adjustable levels of detail based on user needs, adapting its output across multiple dialogue rounds (Table 13), and it can perform detailed writing tasks based on the given image (Table 14).

Table 11. SAISA can generating prompts in a specified format tailored for Stable Diffusion models.

Visual input example, Constrained SD Prompt Generation:	
	
User	Here's a formula for a Stable Diffusion image prompt: a cartoon-style image of [adjective] [subject] [haircut] [clothing] [doing action], [color scheme], detailed, trending on artstation. Generate the prompt for this image.
LLaVA-1.5	A cartoon-style image of a young girl with a heart-shaped bow in her hair, wearing a white and pink jacket, and holding a stuffed bunny. The image has a watercolor color scheme, and it is detailed and trending on ArtStation.
SAISA	A cartoon-style image of a young girl with a cute haircut, wearing a white jacket and a bow, holding a stuffed animal and a backpack. The image has a color scheme of pink, white, and orange, and is detailed and trending on ArtStation.

Table 12. SAISA can evaluate whether the question contains factual inaccuracies and respond to challenging questions.

Visual input example, Tricky Question:	
	
User	If there are factual errors in the questions, point it out; if not, proceed to answering the question. What's happening in the desert?
LLaVA-1.5	There is no desert in the image. The image features a beach with a city skyline in the background, and a large body of water.
SAISA	In the image, there is no desert visible. The scene takes place on a beach with a city skyline in the background. The beach is located near a city, and there is a lively atmosphere with lights reflecting on the water.

Table 13. SAISA is capable of providing responses at various levels of granularity according to user requirements.

Visual input example, Different Format Prompts:

User	What is the color of the shirt that the man is wearing? Answer the question using a single word or phrase.
LLaVA-1.5	Yellow
SAISA	Yellow
User	Is this unusual? Please explain briefly with a short sentence.
LLaVA-1.5	Yes, it is unusual for a man to be drying clothes on the back of a moving vehicle.
SAISA	Yes, it is unusual for a man to be ironing clothes while standing on the back of a taxi.
User	Is this unusual? Please explain in detail.
LLaVA-1.5	Yes, this is an unusual scene. In the image, a man is sitting on a folding chair in the back of a yellow taxi, holding a clothesline with clothes on it. This is not a typical sight in a city setting, as people usually do not transport laundry in such a manner. It is also uncommon to see someone sitting on a folding chair in the back of a taxi, as passengers usually sit inside the vehicle. The combination of the man's unusual activity and the unconventional seating arrangement makes this scene stand out as an atypical and intriguing moment.
SAISA	Yes, this is an unusual scene. In the image, a man is standing on the back of a yellow taxi, holding a clothes iron. This is not a typical sight, as people usually do not use taxis for ironing clothes. It is also not common to see someone ironing clothes in public, as it is generally done indoors in a private setting, such as a home or an office. The man's actions in the image are unconventional and seem to be a humorous or artistic representation of an everyday situation.

Table 14. SAISA is capable of executing complex writing tasks based on a given image.

Visual input example, Visual Writing Task:	
	
User	Compose an engaging travel blog post about a recent trip to this place, highlighting cultural experiences and must-see attractions.
SAISA	Title: Paradise Found: A Trip to the Hawaiian Islands Introduction: I recently had the opportunity to visit the beautiful Hawaiian Islands, and I must say, it was an unforgettable experience. The breathtaking views, crystal blue waters, and lush green landscapes truly make it a paradise on earth. In this blog post, I'll share some of my favorite moments and must-see attractions from my trip. Day 1: Arrival and Exploring Oahu Upon arrival, I was greeted by the stunning view of Waikiki Beach, which is located on the island of Oahu. I spent the day exploring the area, taking in the sights, and soaking up the sun. One of the highlights of my day was visiting the iconic Diamond Head, a volcanic crater that offers panoramic views of the beach and ocean. Day 2: Polynesian Cultural Center The next day, I visited the Polynesian Cultural Center, where I had the chance to experience the rich culture and traditions of various Polynesian islands. The center showcases traditional dances, music, and crafts, as well as interactive exhibits that allowed me to learn more about the history and customs of the region. Day 3: Hanauma Bay On my third day, I visited Hanauma Bay, a stunning natural wonder that is home to an abundance of marine life. The bay is a protected marine sanctuary, and visitors are required to take a guided tour to learn about the local ecosystem and conservation efforts. The crystal blue waters and vibrant coral reefs were truly breathtaking, and I was able to witness sea turtles, colorful fish, and other marine creatures up close. Day 4: Pearl Harbor and USS Arizona On my last day, I visited Pearl Harbor, the site of the infamous attack that led to the United States' entry into World War II. The USS Arizona Memorial is a must-see attraction, as it offers a solemn and educational experience. I was able to learn about the history of the attack and the significance of the USS Arizona in the war. Conclusion: In conclusion, my trip to the Hawaiian Islands was an unforgettable experience filled with cultural experiences, breathtaking views, and adventure. From the iconic Waikiki Beach to the protected marine sanctuary of Hanauma Bay, there is truly something for everyone to enjoy. I would highly recommend a visit to these beautiful islands for anyone looking to experience the wonders of nature and the rich history of the region.